

DESCRIPTION

Trixevas is a round, biconvex, yellow, film-coated tablet marked "T" on one side and plain on the other side.

COMPOSITION

Trixevas Tablet: Each tablet contains 90 mg of ticagrelor.

PHARMACODYNAMICS

Pharmacotherapeutic group: Platelet aggregation inhibitors, ATC code: B01AC24

Mechanism of action

Ticagrelor is a member of the chemical class cyclopentyltriazolopyrimidines (CPTP), which is an oral, direct acting, selective and reversibly binding P2Y₁₂ receptor antagonist that prevents ADP-mediated P2Y₁₂ dependent platelet activation and aggregation. Ticagrelor does not prevent ADP binding, but when bound to the P2Y₁₂ receptor prevents ADP-induced signal transduction. Since platelets participate in the initiation and/or evolution of thrombotic complications of atherosclerotic disease, inhibition of platelet function has been shown to reduce the risk of CV events such as death, MI or stroke.

Ticagrelor also increases local endogenous adenosine levels by inhibiting equilibrative nucleoside transporter-1 (ENT-1). Ticagrelor has been documented to augment the following adenosine-induced effects: vasodilation, inhibition of platelet function and dyspnoea. However, a link between the observed increases in adenosine and clinical outcomes (e.g., morbidity mortality) has not been clearly elucidated.

Pharmacodynamic effects

Onset of action

In patients with stable coronary artery disease (CAD) on ASA, ticagrelor demonstrates a rapid onset of pharmacological effect as demonstrated by a mean inhibition of platelet aggregation (IPA) for ticagrelor at 0.5 hours after 180 mg loading dose of about 41%, with the maximum IPA effect of 89% by 2-4 hours post dose, and maintained between 2-8 hours. 90% of patients had final extent IPA >70% by 2 hours post dose.

Offset of action

If a CABG procedure is planned, ticagrelor bleeding risk is increased compared to clopidogrel when discontinued within less than 96 hours prior to procedure.

Switching data

Switching from clopidogrel 75 mg to ticagrelor 90 mg twice daily results in an absolute IPA increase of 26.4% and switching from ticagrelor to clopidogrel results in an absolute IPA decrease of 24.5%. Patients can be switched from clopidogrel to ticagrelor without any interruption of antiplatelet effect (see section "***DOSAGE AND ADMINISTRATION***").

PHARMACOKINETICS

Absorption

Absorption of ticagrelor is rapid with a median t_{max} of approximately 1.5 hours. The formation of the major circulating metabolite AR-C124910XX (also active) from ticagrelor is rapid with a median t_{max} of approximately 2.5 hours.

The mean absolute bioavailability of ticagrelor was estimated to be 36%. Ingestion of a high-fat meal resulted in a 21% increase in ticagrelor AUC and 22% decrease in the active metabolite C_{max} but had no effect on ticagrelor C_{max} or the AUC of the active metabolite. These small changes are considered of minimal clinical significance; therefore, ticagrelor can be given with or without food. Ticagrelor as well as the active metabolite are P-gp substrates.

Ticagrelor as crushed tablets mixed in water, given orally or administered through a nasogastric tube into the stomach, is bioequivalent to whole tablets with regards to AUC and C_{max} for ticagrelor and the active metabolite. Initial exposure (0.5 and 1 hour post-dose) from crushed ticagrelor tablets mixed in water was higher compared to whole tablets, with a generally identical concentration profile thereafter (2 to 48 hours).

Distribution

The steady state volume of distribution of ticagrelor is 87.5 L. Ticagrelor and the active metabolite is extensively bound to human plasma protein (>99.0%).

Biotransformation

CYP3A4 is the major enzyme responsible for ticagrelor metabolism and the formation of the active metabolite and their interactions with other CYP3A substrates ranges from activation through to inhibition.

The major metabolite of ticagrelor is AR-C124910XX, which is also active as assessed by *in vitro* binding to the platelet P2Y₁₂ ADP-receptor. The systemic exposure to the active metabolite is approximately 30-40% of that obtained for ticagrelor.

Elimination

The primary route of ticagrelor elimination is via hepatic metabolism. When radiolabelled ticagrelor is administered, the mean recovery of

radioactivity is approximately 84% (57.8% in faeces, 26.5% in urine). Recoveries of ticagrelor and the active metabolite in urine were both less than 1% of the dose. The primary route of elimination for the active metabolite is most likely via biliary secretion. The mean $t_{1/2}$ was approximately 7 hours for ticagrelor and 8.5 hours for the active metabolite.

Special populations

Elderly

Higher exposures to ticagrelor (approximately 25% for both C_{max} and AUC) and the active metabolite were observed in elderly (≥ 75 years) ACS patients compared to younger patients by the population pharmacokinetic analysis. These differences are not considered clinically significant (see section (*see DOSAGE AND ADMINISTRATION*)).

Paediatric population

Ticagrelor has not been evaluated in a paediatric population (*see DOSAGE AND ADMINISTRATION and PHARMACODYNAMICS*)

Gender

Higher exposures to ticagrelor and the active metabolite were observed in women compared to men. These differences are not considered clinically significant.

Renal impairment

Exposure to ticagrelor was approximately 20% lower and exposure to the active metabolite was approximately 17 % higher in patients with severe renal impairment (creatinine clearance <30 ml/min) compared to subjects with normal renal function.

In patients with end stage renal disease on haemodialysis AUC and C_{max} of ticagrelor 90 mg administered on a day without dialysis were 38% and 51% higher compared to subjects with normal renal function. A similar increase in exposure was observed when ticagrelor was administered immediately prior to dialysis (49% and 61%, respectively) showing that ticagrelor is not dialysable. Exposure of the active metabolite increased to a lesser extent (AUC 13-14% and C_{max} 17-36%). The inhibition of platelet aggregation (IPA) effect of ticagrelor was independent of dialysis in patients with end stage renal disease and similar to subjects with normal renal function (*see DOSAGE AND ADMINISTRATION*)

Hepatic impairment

C_{max} and AUC for ticagrelor were 12% and 23% higher in patients with mild hepatic impairment compared to matched healthy subjects, respectively, however, the IPA effect of ticagrelor was similar between the two groups. No dose adjustment is needed for patients with mild hepatic impairment. Ticagrelor has not been studied in patients with severe hepatic impairment and there is no pharmacokinetic information in patients with moderate hepatic impairment. In patients that had moderate or severe elevation in one or more liver function tests at baseline, ticagrelor plasma concentrations were on average similar or slightly higher as compared to those without baseline elevations. No dose adjustment is recommended in patients with moderate hepatic impairment (*see DOSAGE AND ADMINISTRATION, WARNING(s) AND PRECAUTION(s)*)

Ethnicity

Patients of Asian descent have a 39% higher mean bioavailability compared to Caucasian patients. Patients self-identified as black had an 18% lower bioavailability of ticagrelor compared to Caucasian patients, in clinical pharmacology studies, the exposure (C_{max} and AUC) to ticagrelor in Japanese subjects was approximately 40% (20% after adjusting for body weight) higher compared to that in Caucasians. The exposure in patients self-identified as Hispanic or Latino was similar to that in Caucasians.

INDICATION(S)

Trixevas co-administered with acetylsalicylic acid (ASA), is indicated for the prevention of atherothrombotic events in adult patients with Acute Coronary Syndromes (unstable angina, non ST elevation Myocardial Infarction [NSTEMI] or ST elevation Myocardial Infarction [STEMI]); including patients managed medically, and those who are managed with percutaneous coronary intervention (PCI) or coronary artery by-pass grafting (CABG)(*see PHARMACODYNAMICS*).

RECOMMENDED DOSAGE

Trixevas treatment should be initiated with a single 180 mg loading dose (two tablets of 90 mg) and then continued at 90 mg twice daily. Patients taking Trixevas should also take ASA daily, unless specifically contraindicated. Following an initial dose of ASA, Trixevas should be used with a maintenance dose of ASA of 75-150 mg (*see PHARMACODYNAMICS*).

Treatment is recommended for up to 12 months unless discontinuation of Trixevas is clinically indicated (*see PHARMACODYNAMICS*). Data beyond 12 months is limited.

In patients with Acute Coronary Syndromes (ACS), premature discontinuation with any antiplatelet therapy, including Trixevas, could result in an increased risk of cardiovascular (CV) death, myocardial infarction (MI) or stroke due to the patient's underlying disease. Therefore, premature discontinuation of treatment should be avoided.

In patients having an ACS event, the loading dose of 180 mg should be given as soon as possible regardless of any previous antiplatelet treatment.

Physicians who, desire to switch patients, with a prior ACS event, to Trixevas should administer the first dose of Trixevas 24 hours following the last dose of the other antiplatelet medication.

Missed dose

Lapses in therapy should be avoided. A patient who misses a dose of Trixevas should take only one 90 mg tablet (their next dose) at its scheduled time.

Special populations

Elderly

No dose adjustment is required in elderly (*see PHARMACOKINETICS*).

Renal impairment

No dose adjustment is necessary for patients with renal impairment (*see PHARMACOKINETICS*).

Hepatic impairment

Severe hepatic impairment - Ticagrelor has not been studied in patients with severe hepatic impairment, hence its use in these patients is contraindicated (*see CONTRAINDICATIONS*).

Moderate hepatic impairment - Only limited information is available. Dose adjustment is not recommended, but ticagrelor should be used with caution (*see WARNING(s) AND PRECAUTION(s) and PHARMACOKINETICS*).

Mild hepatic impairment - No dose adjustment is necessary (*see PHARMACOKINETICS*).

Paediatric patients

The safety and efficacy of ticagrelor in children below the age of 18 has not been established.

METHOD OF ADMINISTRATION: Trixevas can be taken with or without food. For patients who are unable to swallow the tablet(s) whole, the tablets can be crushed to a fine powder and mixed in half a glass of water and drunk immediately. The glass should be rinsed with a further half glass of water and the contents drunk. The mixture can also be administered via a nasogastric tube. It is important to flush the nasogastric tube through with water after administration of the mixture.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients.
- Active pathological bleeding.
- History of intracranial haemorrhage (*see SIDE EFFECTS*).
- Severe hepatic impairment (*see DOSAGE AND ADMINISTRATION, WARNING(s) AND PRECAUTION(s) and PHARMACOKINETICS*)
- Co-administration of ticagrelor with strong CYP3A4 inhibitors (e.g., ketoconazole, clarithromycin, nefazodone, ritonavir and atazanavir), as co-administration may lead to a substantial increase in exposure to ticagrelor (*see DRUG INTERACTIONS*).

WARNINGS AND PRECAUTIONS

Bleeding risk

The use of ticagrelor in patients at known increased risk for bleeding should be balanced against the benefit in terms of prevention of atherothrombotic events (*see SIDE EFFECTS and PHARMACODYNAMICS*).

If clinically indicated, ticagrelor should be used with caution in the following patient groups:

- Patients with a propensity to bleed (e.g., due to recent trauma, recent surgery, coagulation disorders, active or recent gastrointestinal bleeding). The use of ticagrelor is contraindicated in patients with active pathological bleeding, in those with a history of intracranial haemorrhage, and in patients with moderate to severe hepatic impairment (*see CONTRAINDICATIONS*).
- Patients with concomitant administration of medicinal products that may increase the risk of bleeding (e.g. non-steroidal anti-inflammatory drugs (NSAIDs), oral anticoagulants and/or fibrinolytics) within 24 hours of ticagrelor dosing.

Platelet transfusion did not reverse the antiplatelet effect of ticagrelor and is unlikely to be of clinical benefit in patients with bleeding. Since co-administration of ticagrelor with desmopressin did not decrease template-bleeding time, desmopressin is unlikely to be effective in managing clinical bleeding events (*see DRUG INTERACTIONS*).

Antifibrinolytic therapy (aminocaproic acid or tranexamic acid) and/or recombinant factor VIIa may increase haemostasis. Ticagrelor may be resumed after the cause of bleeding has been identified and controlled.

Surgery

Patients should be advised to inform doctors and dentists that they are taking ticagrelor before any surgery is scheduled and before any new medicinal product is taken.

In patients undergoing coronary artery bypass grafting (CABG), ticagrelor had more bleeding than clopidogrel when stopped within 1 day prior to surgery but a similar rate of major bleeds compared to clopidogrel after stopping therapy 2 or more days before surgery (*see SIDE EFFECTS*). If a patient is to undergo elective surgery and antiplatelet effect is not desired, ticagrelor should be discontinued 5 days prior to surgery (*see PHARMACODYNAMICS*).

Patients with prior ischaemic stroke

ACS patients with prior ischaemic stroke can be treated with ticagrelor for up to 12 months.

Treatment beyond one year is not recommended for patients with history of MI with prior ischaemic stroke due to absence of data.

Hepatic impairment

Use of ticagrelor is contraindicated in patients with severe hepatic impairment (*see DOSAGE AND ADMINISTRATION and CONTRAINDICATIONS*).

Caution is advised in patients with moderate hepatic impairment (*see DOSAGE AND ADMINISTRATION and PHARMACOKINETICS*).

Patients at risk for bradycardic events

Holter ECG monitoring has shown an increased frequency of mostly asymptomatic ventricular pauses during treatment with ticagrelor compared with clopidogrel. Ticagrelor should be used with caution in patients with an increased risk of bradycardic events (e.g., patients without a pacemaker who have sick sinus syndrome, 2nd or 3rd degree AV block or bradycardic-related syncope) (*see PHARMACODYNAMICS*).

Caution also should be exercised when administering ticagrelor concomitantly with medicinal products known to induce bradycardia e.g., beta blockers, calcium channel blockers i.e., diltiazem and verapamil, and digoxin (*See DRUG INTERACTIONS*).

Dyspnoea

Dyspnoea was reported in patients treated with ticagrelor. Dyspnoea is usually mild to moderate in intensity and often resolves without need for treatment discontinuation. Patients with asthma/chronic obstructive pulmonary disease (COPD) may have an increased absolute risk of experiencing dyspnoea with ticagrelor. Ticagrelor should be used with caution in patients with history of asthma and/or COPD. The mechanism has not been elucidated. If a patient reports new, prolonged or worsened dyspnoea this should be investigated fully and if not tolerated, treatment with ticagrelor should be stopped. (*see SIDE EFFECTS*).

Creatinine elevations

Creatinine levels may increase during treatment with ticagrelor. The mechanism has not been elucidated. Renal function should be checked according to routine medical practice. In patients with ACS, it is recommended that renal function is also checked one month after initiating the treatment with ticagrelor, paying special attention to patients ≥ 75 years, patients with moderate/severe renal impairment and those receiving concomitant treatment with an angiotensin receptor blocker (ARB).

Uric acid increase

Hyperuricaemia may occur during treatment with ticagrelor (*see SIDE EFFECTS*). Caution is advised in patients with history of hyperuricaemia or gouty arthritis. As a precautionary measure, the use of ticagrelor in patients with uric acid nephropathy is discouraged.

Thrombotic Thrombocytopenic Purpura (TTP)

Thrombotic Thrombocytopenic Purpura (TTP) has been reported very rarely with the use of ticagrelor. It is characterised by thrombocytopenia and microangiopathic haemolytic anaemia associated with either neurological findings, renal dysfunction or fever. TTP is a potentially fatal condition requiring prompt treatment including plasmapheresis.

Interference with platelet function tests to diagnose heparin induced thrombocytopenia (HIT)

In the heparin induced platelet activation (HIPA) test used to diagnose HIT, anti-platelet factor 4/heparin antibodies in patient serum activate platelets of healthy donors in the presence of heparin. False negative results in a platelet function test (to include, but may not be limited to the HIPA test) for HIT have been reported in patients administered ticagrelor. This is related to inhibition of the P2Y₁₂-receptor on the healthy donor platelets in the test by ticagrelor in the patient's sera/plasma. Information on concomitant treatment with ticagrelor is required for interpretation of HIT platelet function tests.

In patients who have developed HIT, the benefit-risk of continued treatment with ticagrelor should be assessed, taking both the prothrombotic state of HIT and the increased risk of bleeding with concomitant anticoagulant and ticagrelor treatment into consideration.

Other

Based on relationship between maintenance ASA dose and relative efficacy of ticagrelor compared to clopidogrel, co-administration of ticagrelor and high maintenance dose ASA (>300 mg) is not recommended (*see PHARMACODYNAMICS*).

INTERACTIONS WITH OTHERS MEDICAMENTS

Ticagrelor is primarily a CYP3A4 substrate and a mild inhibitor of CYP3A4. Ticagrelor is also a P-gp substrate and a weak P-gp inhibitor and may increase the exposure of P-gp substrates.

CYP3A4 inhibitors

- **Strong CYP3A4 inhibitors** – Co-administration of ketoconazole with ticagrelor increased the ticagrelor C_{max} and AUC equal to 2.4-fold and 7.3-fold, respectively. The C_{max} and AUC of the active metabolite were reduced by 89% and 56%, respectively. Other strong inhibitors of CYP3A4 (clarithromycin, nefazodone, ritonavir, and atazanavir) would be expected to have similar effects and therefore concomitant use of strong CYP3A4 inhibitors with ticagrelor is contraindicated (*see CONTRAINDICATIONS*).
- **Moderate CYP3A4 inhibitors** – Co-administration of diltiazem with ticagrelor increased the ticagrelor C_{max} by 69% and AUC to 2.7 fold and decreased the active metabolite C_{max} by 38% and AUC was unchanged. There was no effect of ticagrelor on diltiazem plasma levels. Other moderate CYP3A4 inhibitors (e.g., aprepitant, erythromycin and fluconazole) would be expected to have a similar effect and can as well be co-administered with ticagrelor.

- A 2-fold increase of ticagrelor exposure was observed after daily consumption of large quantities of grapefruit juice (3x200 ml). This magnitude of increased exposure is not expected to be clinically relevant to most patients.

CYP3A inducers

Co-administration of rifampicin with ticagrelor decreased ticagrelor C_{max} and AUC by 73% and 86%, respectively. The C_{max} of the active metabolite was unchanged and the AUC was decreased by 46%, respectively. Other CYP3A inducers (e.g. phenytoin, carbamazepine and phenobarbital) would be expected to decrease the exposure to ticagrelor as well. Co-administration of ticagrelor with potent CYP3A inducers may decrease exposure and efficacy of ticagrelor, therefore, their concomitant use with ticagrelor is discouraged.

Cyclosporine (P-gP and CYP3A inhibitor)

Co-administration of cyclosporine (600 mg) with ticagrelor increased ticagrelor C_{max} and AUC equal to 2.3-fold and 2.8-fold, respectively. The AUC of the active metabolite was increased by 32% and C_{max} was decreased by 15% in the presence of cyclosporine.

No data are available on concomitant use of ticagrelor with other active substances that also are potent P-gp inhibitors and moderate CYP3A4 inhibitors (e.g. verapamil, quinidine) that also may increase ticagrelor exposure. If the association cannot be avoided, their concomitant use should be made with caution.

Others

Co-administration of ticagrelor with heparin, enoxaparin and ASA or desmopressin did not have any effect on the pharmacokinetics of ticagrelor or the active metabolite or on ADP-induced platelet aggregation compared with ticagrelor alone. If clinically indicated, medicinal products that alter haemostasis should be used with caution in combination with ticagrelor.

A delayed and decreased exposure to oral P2Y₁₂ inhibitors, including ticagrelor and its active metabolite, has been observed in patients with ACS treated with morphine (35% reduction in ticagrelor exposure). This interaction may be related to reduced gastrointestinal motility and apply to other opioids. The clinical relevance is unknown, but data indicate the potential for reduced ticagrelor efficacy in patients co-administered ticagrelor and morphine. In patients with ACS, in whom morphine cannot be withheld and fast P2Y₁₂ inhibition is deemed crucial, the use of a parenteral P2Y₁₂ inhibitor may be considered.

Effects of ticagrelor on other medicinal products

Medicinal products metabolised by CYP3A4

- *Simvastatin* – Co-administration of ticagrelor with simvastatin increased simvastatin C_{max} by 81% and AUC by 56% and increased simvastatin acid C_{max} by 64% and AUC by 52% with some individual increases equal to 2- to 3-fold. Co-administration of ticagrelor with doses of simvastatin exceeding 40 mg daily could cause adverse effects of simvastatin and should be weighed against potential benefits. There was no effect of simvastatin on ticagrelor plasma levels. Ticagrelor may have similar effect on lovastatin. The concomitant use of ticagrelor with doses of simvastatin or lovastatin greater than 40 mg is not recommended.
- *Atorvastatin* – Co-administration of atorvastatin and ticagrelor increased atorvastatin acid C_{max} by 23% and AUC by 36%. Similar increases in AUC and C_{max} were observed for all atorvastatin acid metabolites. These increases are not considered clinically significant.
- A similar effect on other statins metabolised by CYP3A4 cannot be excluded.

Ticagrelor is a mild CYP3A4 inhibitor. Co-administration of ticagrelor and CYP3A4 substrates with narrow therapeutic indices (i.e., cisapride or ergot alkaloids) is not recommended, as ticagrelor may increase the exposure to these medicinal products.

P-gp substrates (including digoxin, cyclosporine)

Concomitant administration of ticagrelor increased the digoxin C_{max} by 75% and AUC by 28%. The mean trough digoxin levels were increased about 30% with ticagrelor co-administration with some individual maximum increases to 2-fold. In the presence of digoxin, the C_{max} and AUC of ticagrelor and its active metabolite were not affected. Therefore, appropriate clinical and/or laboratory monitoring is recommended when giving narrow therapeutic index P-gp dependent medicinal products like digoxin concomitantly with ticagrelor.

There was no effect of ticagrelor on cyclosporine blood levels. Effect of ticagrelor on other P-gp substrates has not been studied.

Medicinal products metabolised by CYP2C9

Co-administration of ticagrelor with tolbutamide resulted in no change in the plasma levels of either medicinal product, which suggest that ticagrelor is not a CYP2C9 inhibitor and unlikely to alter the CYP2C9 mediated metabolism of medicinal products like warfarin and tolbutamide.

Oral contraceptives

Co-administration of ticagrelor and levonorgestrel and ethinyl estradiol increased ethinyl estradiol exposure approximately 20% but did not alter the pharmacokinetics of levonorgestrel. No clinically relevant effect on oral contraceptive efficacy is expected when levonorgestrel and ethinyl estradiol are co-administered with ticagrelor.

Medicinal products known to induce bradycardia

Due to observations of mostly asymptomatic ventricular pauses and bradycardia, caution should be exercised when administering ticagrelor concomitantly with medicinal products known to induce bradycardia (*see WARNING(s) AND PRECAUTION(s)*). No evidence of clinically significant adverse reactions after concomitant administration with one or more medicinal products known to induce bradycardia (e.g., beta blockers, calcium channel blockers i.e. diltiazem and verapamil, and digoxin).

Other concomitant therapy

No evidence of clinically significant adverse interactions with ASA, proton pump inhibitors, statins, beta-blockers, angiotensin converting enzyme

(ACE) inhibitors and angiotensin receptor blockers for concomitant conditions for long-term and heparin, low molecular weight heparin and intravenous GpIIb/IIIa inhibitors for short durations (see *PHARMACODYNAMICS*) was seen.

Co-administration of ticagrelor with heparin, enoxaparin or desmopressin had no effect on activated partial thromboplastin time (aPTT), activated coagulation time (ACT) or factor Xa assays. However, due to potential pharmacodynamic interactions, caution should be exercised with the concomitant administration of ticagrelor with medicinal products known to alter haemostasis.

Caution is advised when administering SSRIs (e.g., paroxetine, sertraline and citalopram) with ticagrelor as this may increase the risk of bleeding.

USE IN PREGNANCY AND LACTATION

Women of childbearing potential: Use appropriate contraceptive measures to avoid pregnancy during ticagrelor therapy.

Use in Pregnancy: There are no or limited amount of data from the use of ticagrelor in pregnant women. Studies in animals have shown reproductive toxicity. Ticagrelor is not recommended during pregnancy.

Breast-feeding: Available pharmacodynamic/toxicological data in animals have shown excretion of ticagrelor and its active metabolites in milk. A risk to newborns/infants cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from ticagrelor therapy taking into account the benefit of breastfeeding for the child and the benefit of therapy for the woman.

Fertility: Ticagrelor had no effect on male or female fertility in animals.

SIDE EFFECTS/ADVERSE REACTIONS

Adverse reactions are listed by MedDRA System Organ Class (SOC). Within each SOC the adverse reactions are ranked by frequency categories. Frequency categories are defined according to the following conventions: Very common, common, uncommon, rare, very rare, not known.

Tabulated list of adverse reactions by frequency and system organ class (SOC)

System Organ Class	Frequency	Adverse reactions seen with Ticagrelor
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Uncommon	Tumour bleedings
Blood and lymphatic system disorders	Very Common	Blood disorder bleedings
	Not known	Thrombotic Thrombocytopenic Purpura
Immune system disorders	Uncommon	Hypersensitivity including angioedema
Metabolism and nutrition disorders	Very Common	Hyperuricaemia
	Common	Gout/Gouty arthritis
Psychiatric disorders	Uncommon	Confusion
Nervous system disorders	Common	Dizziness Syncope, Headache
	Uncommon	Intracranial haemorrhage
Eye disorders	Uncommon	Eye haemorrhage
Ear and labyrinth disorders	Common	Vertigo
	Uncommon	Ear haemorrhage
Vascular disorders	Common	Hypotension
Respiratory, thoracic and mediastinal disorders	Very Common	Dyspnoea
	Common	Respiratory system bleedings
Gastrointestinal disorders	Common	Gastrointestinal haemorrhage, Diarrhoea, Nausea, Dyspepsia, Constipation
	Uncommon	Retroperitoneal haemorrhage
Skin and subcutaneous tissue disorders	Common	Subcutaneous or dermal bleeding, Rash, Pruritus
Musculoskeletal and connective tissue disorders	Uncommon	Muscular bleeding
Renal and urinary disorders	Common	Urinary tract bleedings
Reproductive system and breast disorders	Uncommon	Reproductive system bleedings
Investigations	Common	Blood creatinine increased
Injury, poisoning and procedural complications	Common	Post procedural haemorrhage, Traumatic bleedings

SYMPTOMS AND TREATMENT OF OVERDOSE

Ticagrelor is well tolerated in single doses up to 900 mg. Gastrointestinal toxicity is dose-limiting. In the event of overdose, observe for potential adverse reactions such as dyspnoea and ventricular pauses and consider ECG monitoring. There is currently no known antidote to reverse the effects of ticagrelor, and ticagrelor is not dialysable. The expected effect of excessive ticagrelor dosing is prolonged duration of bleeding risk associated with platelet inhibition. Platelet transfusion is unlikely to be of clinical benefit in patients with bleeding. If bleeding occurs appropriate supportive measures should be taken.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Ticagrelor has no or negligible influence on the ability to drive and use machines. During treatment with ticagrelor, dizziness and confusion have been reported. Therefore, patients who experience these symptoms should be cautious while driving or using machines.

STORAGE CONDITIONS

Store in a dry place below 30°C. Protect from light.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACKING/ PACK SIZES

Clear triplex film-alu foil blisters of 3 x 10's and 6 x 10's tablets packed per unit box with a package insert.

All pack sizes may not be marketed.

SHELF LIFE

Please refer to outer product label.

PRODUCT REGISTRATION HOLDER

Duopharma Manufacturing (Bangi) Sdn. Bhd.

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